

```
pip install kagglehub opencv-python tensorflow scikit-learn matplotlib seaborn
```

```
Requirement already satisfied: kagglehub in /usr/local/lib/python3.12/dist-packages (0.3.13)
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (4.13.0.90)
Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (2.19.0)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from kagglehub) (25.0)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from kagglehub) (6.0.3)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from kagglehub) (2.32.4)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from kagglehub) (4.67.1)
Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-python) (2.0.2)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.4.0)
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (0.2.0)
Requirement already satisfied: google-pasta>=0.1.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (18.1.1)
Requirement already satisfied: opt-einsum>=2.3.2 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.4.0)
Requirement already satisfied: protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,<6.0.0dev,>=3.20.3 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (75.2.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from tensorflow) (75.2.0)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing-extensions>=3.6.6 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (4.15.0)
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (2.0.1)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.76.0)
Requirement already satisfied: tensorboard~2.19.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (2.19.0)
Requirement already satisfied: keras>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.10.0)
Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.15.1)
Requirement already satisfied: ml-dtypes<1.0.0,>=0.5.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (0.5.4)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: pandas>=1.2 in /usr/local/lib/python3.12/dist-packages (from seaborn) (2.2.2)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from astunparse>=1.6.0->tensorflow) (0.44.0)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from keras>=3.5.0->tensorflow) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.12/dist-packages (from keras>=3.5.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in /usr/local/lib/python3.12/dist-packages (from keras>=3.5.0->tensorflow) (0.18.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn) (2025.3)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->kagglehub) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->kagglehub) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->kagglehub) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->kagglehub) (2026.1.10)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.12/dist-packages (from tensorboard~2.19.0->tensorflow) (3.7.0)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from tensorboard~2.19.0->tensorflow) (0.7.0)
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from tensorboard~2.19.0->tensorflow) (3.1.0)
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.12/dist-packages (from werkzeug>=1.0.1->tensorboard~2.19.0->tensorflow) (3.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich->keras>=3.5.0->tensorflow) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich->keras>=3.5.0->tensorflow) (2.19.0)
Requirement already satisfied: mdurl~0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich->keras>=3.5.0->tensorflow) (0.1.2)
```

```
import kagglehub, os
```

```
path = kagglehub.dataset_download("shahliza27/ur-fall-detection-dataset")
print("Dataset path:", path)
```

```
video_dir = os.path.join(path, "UR_Fall_Detection_Dataset")
```

Warning: Looks like you're using an outdated `kagglehub` version (installed: 0.3.13), please consider upgrading to the latest version. Downloading from [```
Dataset path: /root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1
```](https://www.kaggle.com/api/v1/datasets/download/shahliza27/ur-fall-detection-dataset?dataset_version_number=1..100%|██████████| 4.18G/4.18G [01:01<00:00, 73.4MB/s]Extracting files...</a></p>
</div>
<div data-bbox=)

```
import cv2, random
import matplotlib.pyplot as plt
```

```
def show_video_sample(video_path):
 cap = cv2.VideoCapture(video_path)
```

```

frames = []
for _ in range(5):
 ret, frame = cap.read()
 if ret:
 frames.append(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))
cap.release()

plt.figure(figsize=(10,2))
for i, f in enumerate(frames):
 plt.subplot(1,5,i+1)
 plt.imshow(f)
 plt.axis("off")
plt.show()

Redefining video_dir to correct the path, assuming videos are directly in the downloaded 'path'
video_dir = path
sample_video = random.choice(os.listdir(video_dir))
show_video_sample(os.path.join(video_dir, sample_video))

```

<Figure size 1000x200 with 0 Axes>

```

import numpy as np

IMG_SIZE = 112
FRAMES = 16

def load_video(path, max_frames=FRAMES):
 cap = cv2.VideoCapture(path)
 frames = []
 while len(frames) < max_frames:
 ret, frame = cap.read()
 if not ret:
 break
 frame = cv2.resize(frame, (IMG_SIZE, IMG_SIZE))
 frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
 frames.append(frame)
 cap.release()

 while len(frames) < max_frames:
 frames.append(frames[-1])

 frames = np.array(frames) / 255.0
 return frames

```

```

import numpy as np
import os
import cv2 # Ensure cv2 is imported for image operations

IMG_SIZE = 112
FRAMES = 16

def load_frames_from_directory(frames_dir, max_frames=FRAMES):
 frames = []
 # List all png files in the directory and sort them by name
 image_files = sorted([f for f in os.listdir(frames_dir) if f.endswith('.png')])

 if not image_files:
 print(f"Warning: No PNG frames found in {frames_dir}. Returning zero-filled placeholder.")
 return np.zeros((max_frames, IMG_SIZE, IMG_SIZE, 3), dtype=np.float32)

 # Strategy: if more than max_frames, take evenly spaced frames. If fewer, duplicate last.
 if len(image_files) > max_frames:
 indices = np.linspace(0, len(image_files) - 1, max_frames, dtype=int)
 selected_image_files = [image_files[i] for i in indices]
 else:
 selected_image_files = image_files

 for img_file in selected_image_files:
 img_path = os.path.join(frames_dir, img_file)
 frame = cv2.imread(img_path)
 if frame is not None:
 frame = cv2.resize(frame, (IMG_SIZE, IMG_SIZE))
 frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
 frames.append(frame)
 else:
 print(f"Warning: Could not read image {img_path}. Skipping.")

```

```

If still fewer frames than max_frames after reading, duplicate the last frame
while len(frames) < max_frames and frames:
 frames.append(frames[-1])

If no frames were ever successfully loaded (e.g., all files unreadable or no PNGs found)
if not frames:
 return np.zeros((max_frames, IMG_SIZE, IMG_SIZE, 3), dtype=np.float32)

frames = np.array(frames, dtype=np.float32) / 255.0
return frames

X, y = [], []

Assuming the structure is video_dir/UR_fall_detection_dataset_cam0_rgb/{video_name}/frames.png
base_dataset_dir = os.path.join(video_dir, "UR_fall_detection_dataset_cam0_rgb")

if not os.path.exists(base_dataset_dir):
 print(f"Error: Base dataset directory not found at {base_dataset_dir}. Please check the dataset structure.")
elif not os.listdir(base_dataset_dir):
 print(f"Error: Base dataset directory {base_dataset_dir} is empty. Please check the dataset structure.")
else:
 for video_folder_name in os.listdir(base_dataset_dir):
 video_folder_path = os.path.join(base_dataset_dir, video_folder_name)
 if os.path.isdir(video_folder_path): # Ensure it's a directory
 label = 1 if "fall" in video_folder_name.lower() else 0
 video_frames = load_frames_from_directory(video_folder_path)
 X.append(video_frames)
 y.append(label)

X = np.array(X)
y = np.array(y)

print("Video tensor shape:", X.shape)

```

Video tensor shape: (70, 16, 112, 112, 3)

```

from sklearn.model_selection import train_test_split

X_train, X_temp, y_train, y_temp = train_test_split(
 X, y, test_size=0.3, stratify=y, random_state=42)

X_val, X_test, y_val, y_test = train_test_split(
 X_temp, y_temp, test_size=0.5, stratify=y_temp)

```

```

for root, dirs, files in os.walk(video_dir):
 for name in files:
 if name.endswith(('.mp4', '.avi', '.mov', '.mkv')):
 print(os.path.join(root, name))
 for name in dirs:
 print(os.path.join(root, name) + '/')

```

```

/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-28-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-14-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-21-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-19-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-12-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-03-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-06-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-26-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-02-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-34-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-04-cam0

```

```

/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-11-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-23-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-04-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-14-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-17-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-10-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-24-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-02-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-30-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-29-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-40-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-07-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-20-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-16-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-39-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-09-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-27-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-32-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-27-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-23-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-01-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-11-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-21-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-06-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-24-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-17-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-15-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-28-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-15-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/fall-10-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-31-cam0
/root/.cache/kagglehub/datasets/shahliza27/ur-fall-detection-dataset/versions/1/UR_fall_detection_dataset_cam0_rgb/adl-08-cam0

```

```

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv3D, MaxPooling3D, Flatten, Dense, Dropout, BatchNormalization
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.metrics import AUC

```

```

model = Sequential([
 Conv3D(32, (3,3,3), activation="relu", input_shape=(FRAMES, IMG_SIZE, IMG_SIZE, 3)),
 BatchNormalization(),
 MaxPooling3D((2,2,2)),

 Conv3D(64, (3,3,3), activation="relu"),
 BatchNormalization(),
 MaxPooling3D((2,2,2)),

 Flatten(),
 Dense(128, activation="relu"),
 Dropout(0.5),
 Dense(1, activation="sigmoid")
])

```

```

/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_shape`
super().__init__(activity_regularizer=activity_regularizer, **kwargs)

```

```

model.compile(
 optimizer=Adam(0.0001),
 loss="binary_crossentropy",
 metrics=["accuracy", AUC(name="auc")]
)

```

```

history = model.fit(
 X_train, y_train,
 validation_data=(X_val, y_val),
 epochs=10,
 batch_size=2
)

```

```

Epoch 1/10
25/25 — 55s 2s/step - accuracy: 0.6701 - auc: 0.7402 - loss: 1.3025 - val_accuracy: 0.7000 - val_auc: 1.0000
Epoch 2/10
25/25 — 80s 2s/step - accuracy: 0.9771 - auc: 0.9737 - loss: 0.3287 - val_accuracy: 1.0000 - val_auc: 1.0000
Epoch 3/10
25/25 — 49s 2s/step - accuracy: 0.9950 - auc: 0.9573 - loss: 0.0832 - val_accuracy: 0.8000 - val_auc: 0.9792
Epoch 4/10
25/25 — 48s 2s/step - accuracy: 1.0000 - auc: 0.9615 - loss: 0.0168 - val_accuracy: 1.0000 - val_auc: 1.0000
Epoch 5/10
25/25 — 52s 2s/step - accuracy: 0.9802 - auc: 1.0000 - loss: 0.0160 - val_accuracy: 0.8000 - val_auc: 0.8958

```

```

Epoch 6/10
25/25 ————— 79s 2s/step - accuracy: 1.0000 - auc: 1.0000 - loss: 1.1272e-08 - val_accuracy: 0.8000 - val_auc: 0.7
Epoch 7/10
25/25 ————— 83s 2s/step - accuracy: 1.0000 - auc: 0.9615 - loss: 7.4845e-06 - val_accuracy: 0.8000 - val_auc: 0.8
Epoch 8/10
25/25 ————— 51s 2s/step - accuracy: 1.0000 - auc: 0.9615 - loss: 0.0075 - val_accuracy: 0.8000 - val_auc: 1.0000
Epoch 9/10
25/25 ————— 82s 2s/step - accuracy: 1.0000 - auc: 1.0000 - loss: 8.2148e-11 - val_accuracy: 0.8000 - val_auc: 1.0
Epoch 10/10
25/25 ————— 49s 2s/step - accuracy: 1.0000 - auc: 0.9615 - loss: 3.0691e-05 - val_accuracy: 0.8000 - val_auc: 1.0

```

```

from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score
import seaborn as sns

```

```

y_prob = model.predict(X_test)
y_pred = (y_prob > 0.5).astype(int)

```

```
print(classification_report(y_test, y_pred))
```

```

cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.show()

```

```
print("AUROC:", roc_auc_score(y_test, y_prob))
```

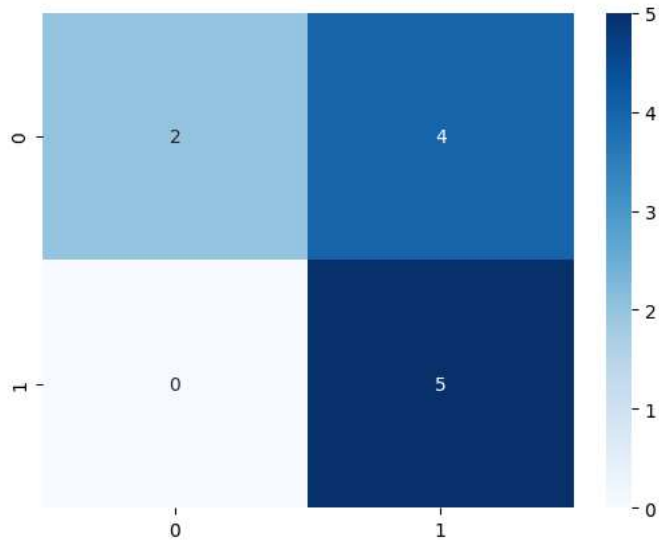
```

1/1 ————— 2s 2s/step
 precision recall f1-score support

 0 1.00 0.33 0.50 6
 1 0.56 1.00 0.71 5

 accuracy 0.78
 macro avg 0.78 0.67 0.61 11
weighted avg 0.80 0.64 0.60 11

```



AUROC: 1.0

